

MDS-UPDRS Finger Tapping Analysis

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [143, 195, 137, 181, 143, 156, 168, 142, 164, 170]

Tap amplitudes (pixels): [157.8131806916013, 174.72549899771354, 151.32745950421557, 168.34488409215172, 151.38031576133008, 153.44380078712857, 153.45683432157722, 136.4734406395618, 135.88230201170424, 118.0889495253472, 140.75865870347016]

GPT-5

MDS-UPDRS Item 3.4 Finger Tapping – Scoring Analysis

I analyzed the provided 10-tap sequence using the tap periods (speed/rhythm) and tap amplitudes (size/decrement). Key points:

1) Tap count and data consistency

- Number of taps reported: 11
- Periods provided: 10 values → these represent the 10 inter-tap intervals for 10 taps.
- Amplitudes provided: 11 values → one amplitude per tap (likely tap 1 through tap 11). For a 10-tap assessment, we evaluate the first 10 taps.

I used the first 10 amplitudes and all 10 periods to score the 10-tap task as instructed.

2) Speed and rhythm (periods)

Tap periods (frames): [143, 195, 137, 181, 143, 156, 168, 142, 164, 170]

- Average period $\approx$ 160 frames with variability.

- Variability pattern: periods alternate between shorter (e.g., 137–156) and longer (e.g., 181–195) intervals. This suggests mild irregularity/slowing but not clear repeated arrests/freezes.
- Interruptions/hesitations: No period is extreme enough to clearly indicate a longer arrest; there are fluctuations, but not obvious >5 interruptions nor a clearly prolonged halt based on the series.

Interpretation for speed/rhythm:

- Consistent with slight-to-mild slowing/irregularity, not moderate (no clear prolonged arrest, and variability isn't severe).

3) Amplitude and decrement

First 10 amplitudes:

[157.8, 174.7, 151.3, 168.3, 151.4, 153.4, 153.5, 136.5, 135.9, 118.1]

- Trend: After a peak at tap 2, amplitudes fluctuate but show a clear downward drift, especially from tap 7 onward.
- Early vs. mid vs. late decrement:
 - Taps 1–3: large amplitudes.
 - Taps 4–6: remain moderately large.
 - Decrement becomes clear by tap 8 and is pronounced by tap 10 (118.1), representing a substantial drop from early taps (~150–175).
- This pattern aligns best with “amplitude decrements midway through the 10-tap sequence” rather than only “near the end” and not “starting after the 1st tap.”

Interpretation for amplitude decrement:

- The decrement emerges around mid-sequence (roughly taps 7–8 onward), matching the “Mild (2)” criterion.

4) Interruptions count

- Based on periods, there are fluctuations but not clearly >5 interruptions; there isn't evidence of a longer arrest. This does not meet “Moderate (3)” a) criteria.
- The variability could account for some hesitations, but not clearly 3–5 distinct interruptions from these data alone.

5) Overall mapping to MDS-UPDRS criteria

- Slight (1): would fit if decrement were only near the end or just slight slowing with 1–2 hesitations.
- Mild (2): matches well due to amplitude decrement becoming evident by the mid-to-late portion of the sequence, with mild slowing/irregularity.
- Moderate (3): not supported—no clear >5 interruptions or arrest; decrement does not start after the 1st tap; speed changes do not appear “moderate.”
- Severe (4): clearly not applicable.

Final judgment: The amplitude behavior is the decisive factor here, and it aligns with "2 (Mild): The amplitude decrements midway through the 10-tap sequence," with speed/rhythm consistent with mild irregularity rather than moderate.

Score: 2
